

Alex Kipnis

PhD Researcher in Bayesian Machine Learning

[Amortized Inference](#) · [Hierarchical Models](#) · [LLM Evaluation](#)

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PhD researcher developing amortized Bayesian inference methods for hierarchical models. Built and released open-source tools for in-context Bayesian mixed-effects regression (`metabeta`) and efficient LLM evaluation (`metabench`). Trained in machine learning, statistics and cognitive neuroscience.

RESEARCH EXPERIENCE

2020 – present

Doctoral Researcher, group of Eric Schulz

Helmholtz Munich (since 2024); MPI for Biological Cybernetics, Tübingen (2020–2023)

- **Amortized Bayesian inference for GLMMs** (`metabeta`; under review, 2026). Neural posterior estimator (set transformers, conditional normalizing flows) that takes data, model and prior specification as inputs, pretrained on millions of simulated datasets and refined by Independence Metropolis–Hastings. It matches NUTS in parameter recovery and calibration on simulated benchmarks and reproduces NUTS posteriors on 40 held-out real datasets, with no dataset-specific retraining or sampler tuning and inference two to three orders of magnitude faster (under 0.1 s per dataset, refinement included). [Open-source PyTorch code](#) and [pretrained weights on Hugging Face](#).
- **Synthetic tabular data prior** (`scamd`). Open-source simulator in the spirit of the TabPFN and TabICL priors: predictor matrices sampled from structural causal models with heterogeneous feature types (continuous, binary, ordinal, count, censored), mixed with columns subsampled from 596 real datasets (PMLB, SRM, UCI). Generates the predictors for `metabeta`'s training data; [released as a standalone Python package](#).
- **Efficient LLM evaluation** (`metabench`; ICLR 2025). Used item response theory to compress six Open LLM Leaderboard benchmarks (about 29k items, more than 5,000 models) to under 3% of their combined items, reconstructing held-out models' scores with an average RMSE across benchmarks of 1.24 points (0–100 scale). [Open-source R code](#); [part of EleutherAI's lm-evaluation-harness](#).

2020

Staff Associate & Master's Thesis, Kriegeskorte Lab

New York

Columbia University

Validated statistical inference methods for representational similarity analysis (RSA).

2019 – 2020

Research Intern & Research Assistant

MPI for Human Cognitive and Brain Sciences, Leipzig; Biological Psychology, LMU Munich

SELECTED PUBLICATIONS & PREPRINTS

2026

Prior-Amortized In-Context Bayesian Inference for Generalized Linear Mixed-Effects Models. *Under review.* [arXiv:2609.24422v1](https://arxiv.org/abs/2609.24422v1)

Alex Kipnis, Marcel Binz, and Eric Schulz.

2025

A sparse benchmark of reasoning and knowledge in Large Language Models. *ICLR 2025.* [OpenReview](#)

Alex Kipnis, Konstantinos Voudouris, Luca M. Schulze Buschoff, and Eric Schulz.

2023

Statistical inference on representational geometries. *eLife.* [doi:10.7554/eLife.82566](https://doi.org/10.7554/eLife.82566)

Heiko Schütt, Alex Kipnis, Jörn Diedrichsen, and Nikolaus Kriegeskorte.

EDUCATION

2020 – present	Doctoral studies in machine learning (in progress) Helmholtz Munich · supervisor Eric Schulz
2018 – 2020	M.Sc. Neuro-Cognitive Psychology LMU Munich · grade 0.9 · thesis at Columbia University
2015 – 2018	B.Sc. Psychology Heidelberg University · grade 1.2
2020 – 2023	Additional coursework in Statistics, Machine Learning & Computer Science Graduate coursework in statistics and machine learning (LMU Munich and University of Tübingen; 42 ECTS, grade 1.1); undergraduate coursework in statistics and computer science (LMU Munich; 81 ECTS, grade 1.2).

TECHNICAL SKILLS

Programming	Python (PyTorch, NumPy, PyMC), R, Bash, Git.
Engineering	Unit-tested, type-annotated research code; experiment tracking with Weights & Biases; training and evaluation on SLURM clusters; model release on Hugging Face.
Methods	Amortized and simulation-based inference (normalizing flows, set transformers); MCMC (NUTS, Independence Metropolis–Hastings); hierarchical Bayesian models and posterior evaluation (recovery, coverage, simulation-based calibration); structural causal models; Gaussian processes; item response theory and factor analysis.
Open source	metabeta (PyTorch), scamd (Python), metabench (R); contributor to rsatoolbox and The Book of Statistical Proofs .
Languages	English, German, Russian.

SERVICE, PRESENTATIONS & AWARDS

2026	Reviewer: ICLR 2026 (RSI Workshop); NeurIPS 2026
2025 – 2026	Posters: metabench (ICLR 2025), metabeta (HAICON 2026)
2023 – 2026	PhD scholarship, German Academic Scholarship Foundation (Studienstiftung)
2020	DAAD scholarship for the master's thesis stay at Columbia University
2018 – present	Member, Elite Network of Bavaria (state programme for top-performing students)